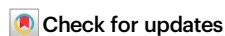


Transport chirality generated by a tunable tilt of Weyl nodes in a van der Waals topological magnet

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Chirality – a characteristic handedness that distinguishes ‘left’ from ‘right’ – is a fundamental property of quantum particles under broken symmetry intimately connected to their spins. Chiral fermions have been identified in Weyl semimetals through their unique electrodynamics arising from ‘axial’ charge imbalance between pairs of chiral Weyl nodes – the topologically protected ‘relativistic’ crossings of electronic bands. Chiral magnetotransport phenomena critically depend on the details of electronic band structure. However, the putative emergence of chiral electronic channels through shape altering of Weyl nodes is yet to be revealed. Here, we detect chirality-endowed linear conduction channels promoted by a tilt of Weyl bands in inversion-symmetric Weyl ferromagnet MnSb₂Te₄. The tuning of Weyl nodes is controlled with ionic hydrogen, which heals the system’s (Mn-Te) bond disorder and lowers the internode scattering. The reshaped Weyl states feature a doubled Curie temperature ≥ 50 K and a strong angular transport chirality synchronous with a rare field-antisymmetric longitudinal resistance – a low-field tunable ‘chiral switch’ that is rooted in the interplay of Berry curvature, chiral anomaly and a hydrogen-mediated form of Weyl nodes.

Chiral Weyl fermions – spin 1/2 massless particles introduced in relativistic field theories¹ but never found as elementary particles in nature – have been predicted to emerge as low-energy excitations in condensed matter². Discovered only recently³, the Weyl quasiparticles have since been identified in the so-named Weyl semimetals^{4,5}, a three-dimensional (3D) gapless topologically nontrivial phases with protected linear two-band crossings in the Brillouin zone of the reciprocal space. When either time reversal ($\mathcal{T}$) or inversion ($\mathcal{I}$) symmetries are broken, the crossing points (Weyl nodes) are doubly degenerate and come in pairs with node partners having an opposite chirality set by the parallel or antiparallel locking of quasiparticle spin to its momentum. The Weyl nodes are topologically robust (the two nodes cannot be

trivially gapped out) because they carry a topological charge, a quantized Berry flux through any surface enclosing them in momentum space⁴. This charge, an integer topological invariant $C = \pm 1$ called the Chern number, gives the electronic wavefunction a handedness. Weyl nodes act as a sink or source of the Berry curvature⁶, which in turn acts as a magnetic field in the momentum space. All this has a huge impact on the electrodynamics of Weyl fermions, which have no counterpart in conventional metals with phenomena such as the chiral anomaly – the nonconservation of the chiral charge deriving from a right–left imbalance between the occupancies $n_{R/L}^{3D}$ of the gapless, zeroth Landau levels with opposite chiralities⁴ under parallel electric and magnetic fields, namely $\frac{dn_{R/L}^{3D}}{dt} = \pm \frac{e^2}{h^2} \mathbf{E} \cdot \mathbf{B}$ relevant to transport in 3D.

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Although various Weyl and Weyl-like phases have been widely reported, including Type II Weyl semimetals^{7,8}, hybrid and higher-order semimetals^{9,10}, and magnetic Weyl semimetals^{11,12}, current materials classification schemes¹³ are yet to provide a clear guiding principle for dialing in chiral Weyl fermion dynamics prescribed by the specific form of Weyl nodes^{14,15}. This is particularly challenging in magnetic ($\mathcal{T}$ -symmetry broken) systems where spin textures, by exerting “axial gauge fields”¹⁶ on Weyl fermions, may reorganize their collective behaviors¹⁷ in unexpected ways.

In this work, we report the discovery of hydrogen-tunable topological bandstructures that generate enhanced and accessible chiral transport of charge in the ferromagnetic (FM) Weyl state. The transformed Weyl semimetal phases are realized by tapping into the existing disorder landscape in the FM MnSb_2Te_4 using the post-growth hydrogen intake-and-release technique we have recently developed¹⁸, which turns Type I Weyl semimetal⁴ with moderately tilted Weyl nodes into a more strongly tilted and rotated Weyl-node system. The modified Weyl FM state displays strong field antisymmetry of the chiral-anomaly driven transport¹⁹ where enhanced angular chirality and reduced internode scattering under in-plane magnetic fields translates into out-of-plane field-antisymmetric chiral currents²⁰, massively amplified at low fields. The role of ionic hydrogen is two-fold: through hybridization with Te, it alters magnetic anisotropy and, by healing the ever-present dangling Mn-Te bonds, enforces robust and reproducible transport asymmetries driven by Berry curvature^{6,21}. The latter puts into sharp relief the key role of bond/site disorder in controlling topological phases and shows a path to a Weyl bandstructure yielding unconventional real-space chiral charge transport by the defect-altering of available materials with intrinsic phase space chirality.

A superabundance of exotic magnetic topological quantum states was recently predicted²² in an important XP_2C_4 class of $\mathcal{T}$ -symmetric topological magnets (X = transition-metal or rare-earth element, P = group 15 pnictogens Bi or Sb, C = group 6A chalcogens Te, Se, or S), which now include a newly discovered intrinsic axion²³ and quantum anomalous Hall (QAH) insulators^{24,25} in MnBi_2Te_4 , and a Type II Weyl semimetal with one pair of Weyl nodes in antiferromagnetic (AFM) $\text{Mn}(\text{Bi}_{1-x}\text{Sb}_x)_2\text{Te}_4$ when Bi is substituted by Sb^{26–28}. In XP_2C_4 , magnetic X elements ideally are all arranged as a 2D ferromagnetic sheet in a layered van der Waals (vdW) crystal structure consisting of C-P-C-X-C-P-C septuple layers (SLs), separated by vdW gaps yet coupled antiferromagnetically²⁹. In $\text{Mn}(\text{Bi}_{1-x}\text{Sb}_x)_2\text{Te}_4$, owing to the comparable sizes of Mn and Sb, it is easy to form Sb_{Mn} antisite (interchange) defects^{30,31}, which can affect the interaction between SLs and turn the long-range order from AFM to FM. Changes in the magnetic structure have been shown to influence the topology of the low energy bands^{28,30,31}, and the particular bandstructure of FM MnSb_2Te_4 was found³² to be Weyl Type I, as long as the system is not Mn-rich.

Results

Here we ‘redesign’ a metallic-like nominally MnSb_2Te_4 (MST) single crystal with the uniform septuple layer (SL) crystal structure (Supplementary Figs. 1, 2), grown as p -type (hole) conductor (Supplementary Figs. 3–5) typical of Sb-based systems²⁶. It contains ample antisite disorder (Supplementary Fig. 1c) and is slightly off-nominal stoichiometry (see the “Methods” section), as requisite for the FM order³⁰. The mean-field-like net magnetization $M(T)$, set here by a joint action of SLs and the antisite-controlled inter-SL coupling, aligns out-of-plane and onsets at the Curie temperature $T_C \cong 26$ K, in correspondence with the characteristic cusp in the temperature dependence of longitudinal resistance $R_{xx}(T)$ (Fig. 1a), all consistent with prior reports³⁰.

Transforming magnetic order of MnSb_2Te_4 with hydrogen

Our first immediately notable and surprising result is that insertion of ionic hydrogen (Fig. 1b) converts the as-grown FM MST into an antiferromagnet, which is seen in the typical AFM signatures in

magnetization $M(T)$ and $R_{xx}(T)$ (Fig. 1c) as well as in the transverse (Hall) signal $R_{yx}(T)$ which tracks the peak in $M(T)$ at the Néel temperature $T_N \approx 20$ K, comparable to as-grown AFM MST³⁰. Here, the level of hydrogenation was controlled by timing the diffusion of H^+ ions from a dilute (0.5 molar) $\text{HCl} + \text{H}_2\text{O} = \text{H}^+(\text{H}_2\text{O}) + \text{Cl}^-$ solution maintained at room temperature in which the samples were immersed for a period of minutes to hours (see the “Methods” section)—a process invented and used by us previously¹⁸ to reach charge neutrality and topological surface states through in-diffusion of H^+ into the bulk. In MST, hydrogen intake also reduces carrier (hole) density in the AFM state (Supplementary Figs. 5, 6 and Supplementary Note 7) by donating electrons to the system, although the process appears self-limiting, and the MST system remains p -type throughout.

A reverse process—the release of hydrogen (Fig. 1e) by a low-temperature annealing protocol¹⁸—returns the system to an FM state, however, this new FM (FM 2) differs from the as-grown one (FM 1) in two spectacular ways: (i) the T_C can be nearly doubled up to ≈ 54 K (Fig. 1f and Supplementary Fig. 6) and (ii) the longitudinal resistance turns from field-symmetric, i.e. $R_{xx}(H) = R_{xx}(-H)$ in FM 1 to field-antisymmetric $R_{xx}(H) = -R_{xx}(-H)$ in FM 2 (Fig. 1g, i). We remark that annealing of as-grown MST leaves T_C and the field-symmetric $R_{xx}(H)$ intact (Supplementary Fig. 8).

Field-antisymmetric longitudinal resistance in MnSb_2Te_4

To unpack the origins of this transformation, cued by our magnetization measurements (Supplementary Fig. 9), we begin by asking whether (and how) the hydrogen intake-release process would affect magnetic anisotropy. From X-ray photoelectron spectroscopy (see the “Methods” section), we determine that hydrogen mainly binds to Mn on the vacant Sb sites (Sb_{Mn} antisites) and that after hydrogenation cycles, the broken Mn-Te bonds (abundant in as-grown MST) are passivated (Supplementary Fig. 10). Our ab initio (DFT) calculations confirm the formation of Mn-H-Te moieties (Supplementary Figs. 11, 12 and Supplementary Table 1) in the FM 2 state, which leads to doubling of T_C and indicates that tilting magnetic easy axis towards normal to the Mn-H bonding direction lowers the energy of the ground state (Supplementary Note 7, Supplementary Table 2).

We test this further by first examining angular dependencies of the Hall signals under tilt θ of the magnetic field (see cartoon in Fig. 2) in FM 1 and FM 2 (Fig. 2a, b). In FM 1, the preferred magnetic alignment is along the crystalline c -axis, while in FM 2, the 45° incline of the preferred direction is evident, albeit notably, it appears only at low fields in the magnetization reversal range where magnetic domains proliferate. We note that the hydrogen-modified easy axis (Fig. 2c, d) could impact the average domain wall direction, although, as we will show later with the low-field in-plane transport, it could also affect Weyl nodes. This is consistent with the longitudinal and Hall signals in FM 1 (Fig. 2e, f, Supplementary Fig. 13) and FM 2 (Fig. 2g, h, Supplementary Fig. 14) that clearly show the c -axis alignment in FM 1 and a remarkable asymmetry at low fields in FM 2—a split in the $R_{xx}(\theta)$ into a dip (d) at $\theta = 45^\circ$ and a peak (p) slightly above $\theta = 90^\circ$ when magnetic field is grazing the ab -plane. This asymmetry is seconded in $R_{yx}(\theta)$ below 50 mT as a kink-like shift to a distinct 45° angle. We point out that at higher fields, the c -axis alignment in transport is entirely restored (Fig. 2h, Supplementary Fig. 14).

Field-antisymmetric magnetoresistance and planar Hall effect

For the field fully tilted to in-plane, a huge field-antisymmetric transverse signal $R_{yx}(H)$ dubbed Planar Hall effect¹⁹ or PHE is detected in both FM 1 and FM 2 (Fig. 2i, j, also Supplementary Figs. 15–18). The observed planar antisymmetry is uncommon—it has been detected in Weyl systems with antisymmetric magnetization and the outwardly tilted Weyl cones²⁰, and it was recently shown³³ that Weyl tilt will lead to an antisymmetric response of chiral anomaly-induced longitudinal magnetoconductivity $\sigma_{xx}(H)$, as well as in chiral anomaly-induced

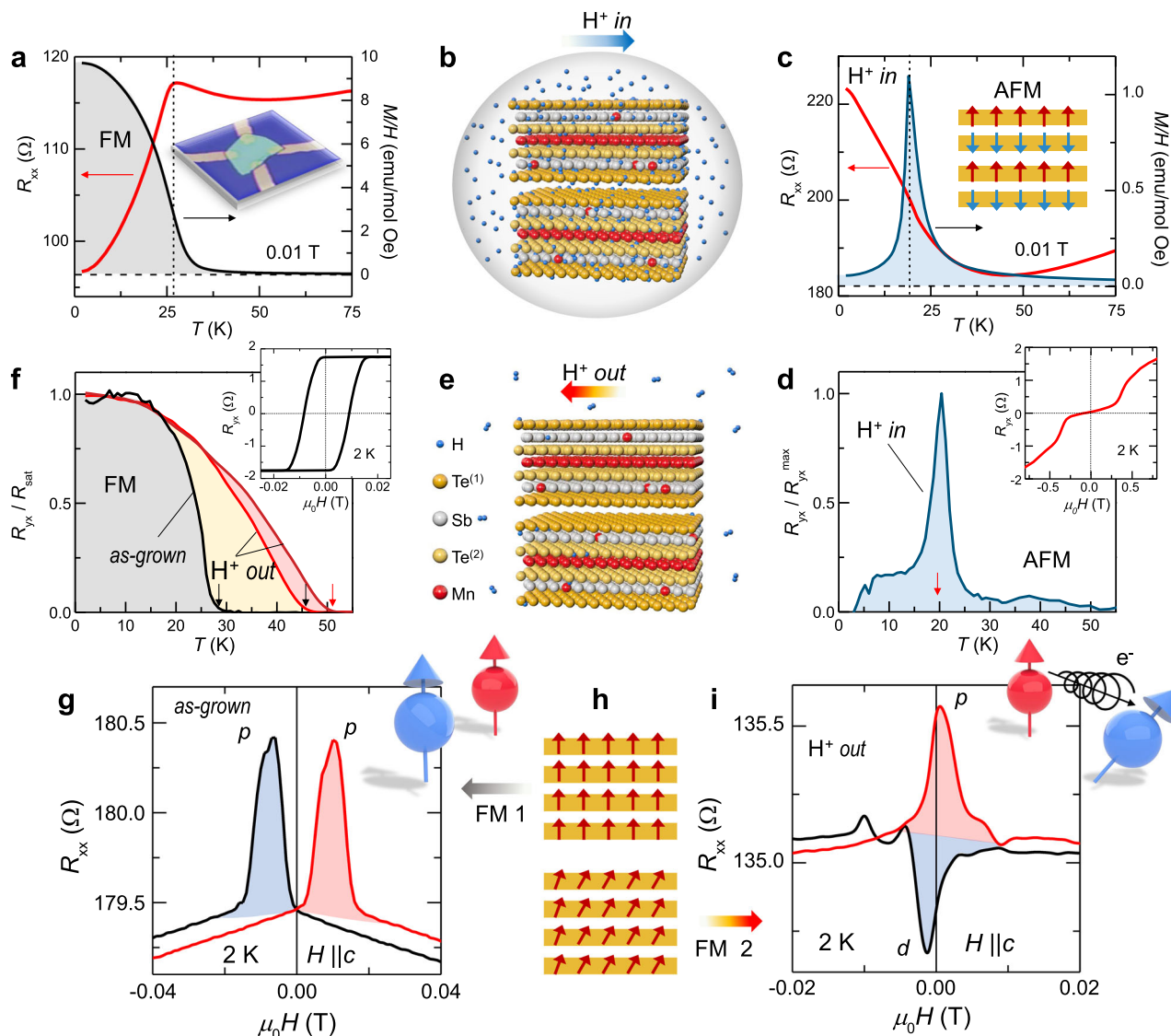


Fig. 1 | Hydrogen-induced transitions between disparate spin orders in MnSb_2Te_4 . **a** Longitudinal resistance R_{xx} and magnetization M vs. temperature of as-grown ferromagnetic MnSb_2Te_4 (MST). The cusp in $R_{xx}(T)$ at the Curie temperature $T_C \approx 26$ K coincides with the maximum slope in $M(T)$. The field is out-of-plane ($\parallel c$ -axis). *Inset*: Optical image of the van der Pauw contact geometry used. **b** Illustration of hydrogenation: the sample is submerged in a dilute aqueous HCl solution (0.5 M) at room temperature, where H^+ permeates the bulk through a timed diffusion process. **c** Hydrogenation turns the as-grown MST (FM 1) into a robust antiferromagnet (AFM) (also Supplementary Fig. 6) with Néel temperature $T_N \approx 20$ K, akin to as-grown AFM MST. A typical AFM behavior is evident in $R_{xx}(T)$ and $M(T)$ (see the “Methods” section). **d** Transverse (Hall) signal $R_{yx}(T)$ tracks AFM

magnetization in (c). *Inset*: A characteristic AFM quasi-plateau²⁹ in $R_{yx}(H)$ after hydrogenation. **e** Illustration of dehydrogenation: H^+ inside the sample is released as H_2 gas by an anneal in the 100–130 °C temperature range prescribed by the sample thickness. **f** Temperature evolution of the Hall resistance R_{yx} after the release of hydrogen. This returns the system to a ferromagnetic state (FM 2) with a nearly doubled T_C . *Inset*: R_{yx} hysteresis loop in as-grown MST. **g** H -symmetric longitudinal resistance R_{xx} with two peaks (p - p) at the magnetization reversal in the FM 1 state. *Inset*: Illustration of the H -symmetric magnetoresistance (MR) in FM 1. **h** Schematic magnetic alignments in FM 1 and FM 2. **i** $R_{xx}(H)$ in FM 2 is H -antisymmetric; it has flipped into a dip and a peak (d - p). *Inset*: Illustration of the chiral H -antisymmetric MR in FM 2.

planar Hall conductivity $\sigma_{yx}(H)$ (see Supplementary Note 20). Notably, an unusual in-plane anomalous $R_{yx}(H)$ loop (ARL) centered at $H=0$ appears in both; it is tiny in FM 1 but is super-pronounced in FM 2. Such ARL loops with $H \parallel ab$ are rare; it is a linear effect (distinct from PHE) that emerges from an out-of-plane Berry curvature³⁴ when the net magnetization is off the symmetry axis²⁰. Notably, in FM 2, the magnetic easy axis is in the plane normal to the Mn-H bond direction (Supplementary Table 2). Both the PHE and the ARL disappear at the respective T_C s of FM 1 and FM 2 (Fig. 2i, j). We point out that in FM 1, where two field-symmetric peaks (p - p) are seen in $R_{xx}(H \parallel c)$ the minuscule ARL with H_c^{ab} vanishes at T_C . In FM 2, on the other hand, H_c^{ab} is by comparison immense in the T -range where $R_{xx}(H \parallel c)$ is H -antisymmetric, i.e. featuring a peak and a dip (p - d) seen in Fig. 1i. In FM

2, H_c^{ab} becomes nearly null at $T_{dp} \approx 40$ K before reentering at the level of FM 1 (Fig. 2k), where the symmetric p - p behavior is also recovered. The resistance difference between the peak and the dip, $\Delta^{p-d}R_{xx}(T)$ (see Fig. 2l), which switches to symmetric p - p at T_{dp} (also Supplementary Figs. 15, 16), confirms the correlation of both PHE and ARL with the antisymmetric p - d .

The most striking visual of the connection between the peak and dip and the antisymmetry of PHE is obtained from its evolution with the field inclination (from the c -axis) angle θ . The p - d field coordinates (Fig. 2m) are very near zero fields at first, but then split at the aforementioned angle $\theta = 45^\circ$ and find a terminus at the antisymmetric ‘wings’ of PHE when $\theta = 90^\circ$ ($H \parallel ab$) (see Fig. 2m, n). This connection is reproducible and robust; it is also

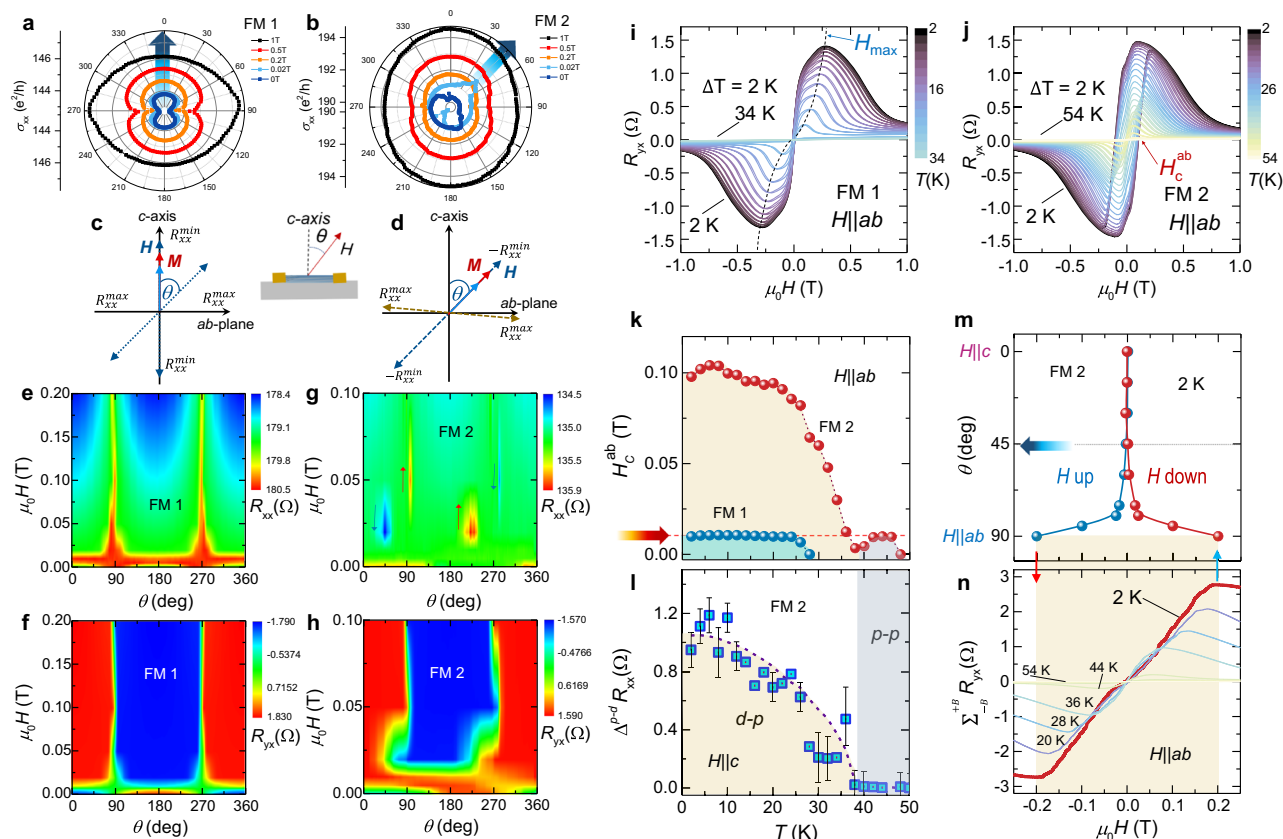


Fig. 2 | Hydrogen-induced transport anisotropy of ferromagnetic MnSb_2Te_4 under tilted magnetic fields. **a, b** Longitudinal conductance σ_{xx} under the field tilt in the FM 1 and FM 2 states, respectively. Polar plots of σ_{xx} show magneto-conductance favoring the c -axis in FM 1, while in FM 2, the preferred axis of 45° appears at low fields. **c, d** Illustration of the relative low-field alignments of magnetic field H , magnetization M , and the average direction of domain walls in the FM 1 and FM 2 states, respectively. **e, f** Low-field 3D contour plots of R_{xx} and R_{yx} in the H - θ plane for the FM 1 state. **g, h** Low-field 3D contour plots of R_{xx} and R_{yx} in the FM 2 state. In FM 1, the resistance is maximum when the field is in-plane ($\theta = 90^\circ$). In FM 2, a ‘negative’ R_{xx} signal (blue arrow down) is detected at $\theta = 45^\circ$, and a ‘positive’ signal (red arrow up) follows (dip and peak). The response now is antisymmetric across 180° . **h** The low-field asymmetry of FM 2 shows up in the Hall signal R_{yx} in the

H - θ plane. The in-plane transverse (Planar Hall) signal $R_{yx}(H)$ —in **i**, for FM 1 and **j**, for FM 2 (see text). A zero-field-centered hysteresis loop with a width H_c^{ab} is tiny in FM 1 but becomes large in FM 2. **k** Temperature evolution of H_c^{ab} . In FM 2, H_c^{ab} nearly vanishes at $T_{\text{dp}} \sim 40$ K, but then reenters at the level of FM 1 to vanish at T_C . **l** $\Delta^{p-d} R_{xx}$, the resistance difference between the dip and the peak, disappears at T_{dp} at which the H -symmetric R_{xx} is recovered. The error bars in Δ^{p-d} indicate experimental uncertainty in the positions of peak and dip. **m, n** A direct connection between H -antisymmetric R_{xx} and the PHE signal $R_{yx}(H)$. **m** Evolution of the peak and dip (p - d) fields with the tilt angle θ at 2 K. **n** For $H \parallel ab$, d - p transforms into the ‘wings’ of the symmetrized PHE = $\sum_{-B}^B R_{yx}$, a sum taken over up- and down-field sweeps, which eliminates the contribution from the in-plane Hall loop.

found after a repeat hydrogenation cycle stabilizing the hydrogen-transformed FM state (Supplementary Fig. 18). The progressive tendency of the subsequent hydrogenation cycles, namely the increase in T_C and in the width H_c^{ab} of the in-plane loop, as well as the flip of the p - d order with a corresponding flip of the loop circulation (Supplementary Figs. 18, 19) highlight the apparent tunability of the process.

Angular variation of the in-plane-field transport

From the above, we surmise that the in-plane transport holds the key to hydrogen-induced H -antisymmetric $R_{xx}(H)$ in MST for $H \parallel c$. In service of this idea, we next examine and compare transport parameters under in-plane (φ -angle) field rotation (see cartoon in Fig. 3) in the FM 1 and FM 2 states. We immediately note the salient distinctions in the in-plane field dependencies: in FM 1, the longitudinal magnetoconductance (MC) $\sigma_{xx}(H)$ is H -parabolic ($\propto H^2$), while it is strictly linear in FM 2, excluding the low-field (PHE-related) hysteretic region (Fig. 3a, b). In theory^{14,35} these MC behaviors have been associated with tilted Type I and Type II Weyl nodes, respectively. We find that such field dependencies are also obtained by tuning the internode/intranode scattering strength (Supplementary Note 20, Supplementary Figs. 17, 18). Here, the difference is also striking at low (≤ 250 mT) fields, displaying

symmetric MC in FM 1 and antisymmetric one in FM 2, the latter likely reflecting the impact of tilting that would allow the contributions from both, PHE and ARL³³. The dissimilarity is further notable in the 3D plots of $\sigma_{yx}(H, \varphi)$ —in contrast to FM 1 (Fig. 3c), in FM 2 σ_{yx} changes polarity at $\varphi = 90^\circ$ (Fig. 3d). For another perspective of the 3D plots see Supplementary Fig. 20.

Our recorded φ variations of the in-plane transverse and longitudinal conductances, $\sigma_{yx}(\varphi) = \Delta\sigma \sin(\varphi + \varphi_0)$ and $\Delta\sigma_{xx}^{\text{ave}}(\varphi) = \sigma_{\perp} + \Delta\sigma \cos^2(\varphi + \varphi_0)$, are as anticipated in a Weyl system³⁵; they globally match up in FM 1 (Fig. 3e) and FM 2 (Fig. 3g), albeit in FM 2 σ_{yx} is phase-shifted by 90° relative to FM 1 (see Supplementary Note 20). Here σ_{yx} is the planar Hall signal (PHE), and $\Delta\sigma = |\sigma_{\perp} - \sigma_{\parallel}|$ is the anisotropy in conductivity due to chiral anomaly arising from the $\mathbf{E} \cdot \mathbf{B}$ term, and thus expected to be maximal for $\sigma_{\parallel} = \sigma_{xx}(\varphi = 0^\circ)$ and minimal for $\sigma_{\perp} = \sigma_{xx}(\varphi = 90^\circ)$. We note that PHE is a more reliable measure of a chiral anomaly than a positive MC in Fig. 3a, b, which crucially depends on disorder modification of the inter- and intranode scattering events³⁶.

Hydrogen-mediated tilt and rotation of Weyl nodes

In what follows, we show that $\sigma_{yx}(\varphi)$ distortions in FM 2 are associated with the tilt mutation of Weyl nodes. We begin by noting that

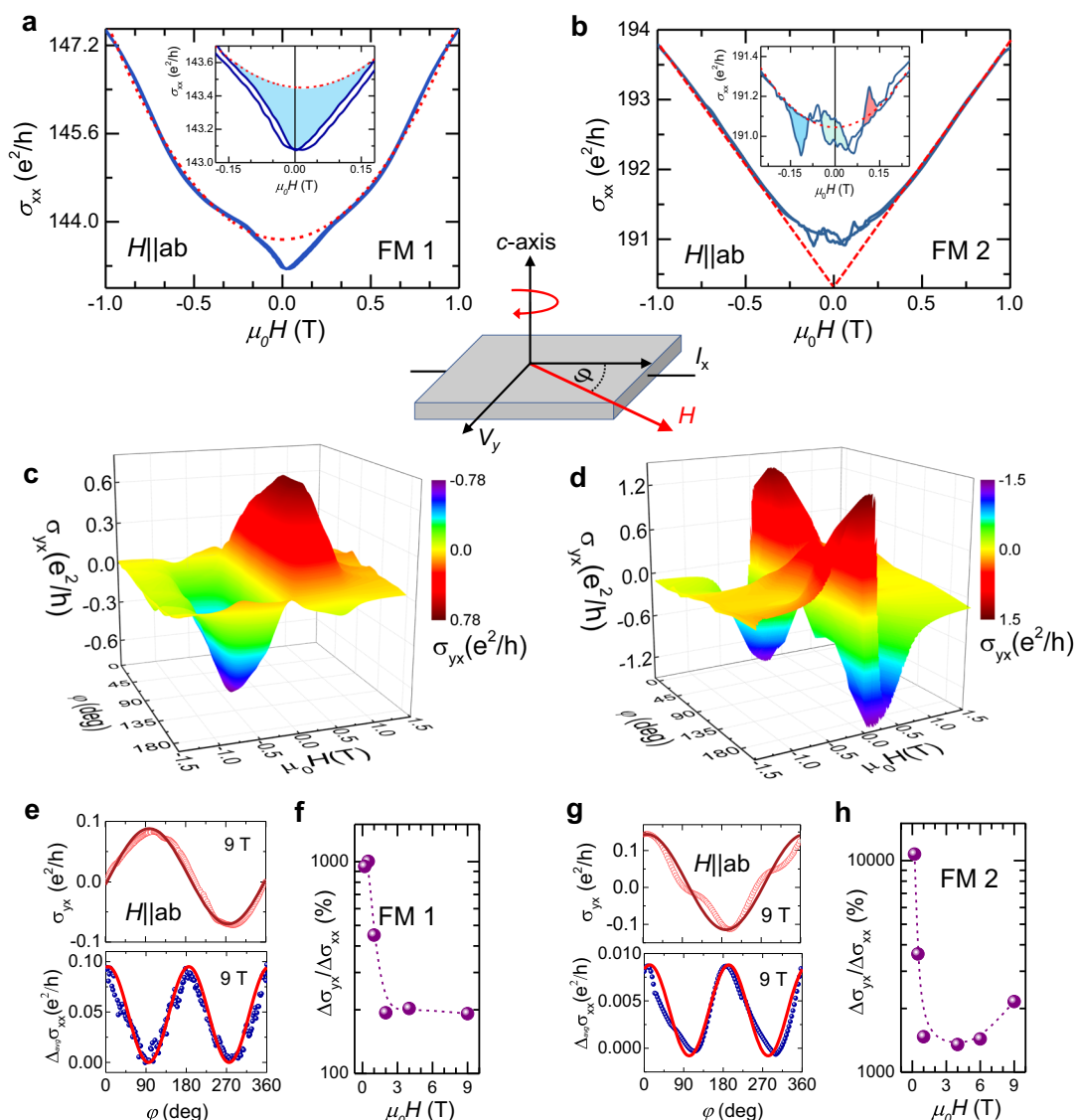


Fig. 3 | In-plane charge transport in the as-grown and hydrogen-modified FM states of MnSb_2Te_4 . **a** Field dependence of the longitudinal conductance σ_{xx} in as-grown ferromagnetic MnSb_2Te_4 (FM 1) at 2 K. Beyond 200 mT, $\sigma_{xx}(H)$ is nearly quadratic in H . *Inset*: In the magnetization reversal region at low fields ($H < 200$ mT) $\sigma_{xx}(H)$ is quasilinear. **b** Field dependence of σ_{xx} in hydrogen-modified ferromagnetic MnSb_2Te_4 (FM 2) at 2 K. Beyond 200 mT, $\sigma_{xx}(H)$ is linear in H . *Inset*: The antisymmetric MC at low fields now comprises the center hysteresis loop and dip and peak features originating from the PHE signal shown in Fig. 2j. **c** 3D plot of the PHE conductance $\sigma_{yx}(H, \varphi)$ in FM 1 over a large field and angle range. **d** 3D plot of the PHE conductance $\sigma_{yx}(H, \varphi)$ in FM 2 over a large field and angle range shows it to be field-antisymmetric about $H = 0$ (see text). Plan-view projections of these 3D plots are

shown in Supplementary Fig. 20. **e** In-plane angular (φ) dependence of transport in FM 1 at 9 T obtained by rotating field in the plane (see cartoon). Top: Planar Hall effect (PHE) $\sigma_{yx}(\varphi) \propto \sin \varphi$. Bottom: $\Delta \sigma_{xx}^{\text{avg}}(\varphi) \propto \cos^2 \varphi$, the φ -change in field-averaged longitudinal magnetoconductance (MC) $\sigma_{xx}^{\text{avg}} = \frac{1}{2}(\sigma_{xx}(H) + \sigma_{xx}(-H))$ with the background subtracted for each value of H . The planar Hall signal (PHE) in FM 1 is large; the ratio of PHE to the change in MC reaches 1000% at low fields. **g** The in-plane angular dependencies of $\sigma_{yx}(\varphi)$ (top) and $\Delta \sigma_{xx}^{\text{avg}}(\varphi)$ (bottom) in FM 2. Here, the deviations of $\sigma_{yx}(\varphi)$ from $\cos \varphi$ are clearly evident (see angular decomposition analysis in Supplementary Fig. 21). **h** PHE signal in FM 2 is further enhanced by an order of magnitude.

$\mathcal{R} = \frac{\Delta \sigma_{yx}(\varphi)}{\Delta \sigma_{xx}^{\text{avg}}(\varphi)}$ —the ratio of φ -oscillation amplitudes in σ_{yx} to σ_{xx} —is exceptionally large in FM 1 (Fig. 3f) and truly giant in FM 2 (Fig. 3h), as was predicted for PHE in a Weyl semimetal¹⁹. Here, $\Delta \sigma_{yx} = |\sigma_{yx}^{\text{max}} - \sigma_{yx}^{\text{min}}|$, $\Delta \sigma_{xx}^{\text{avg}} = \frac{1}{2}|\sigma_{xx}^{\text{max}} - \sigma_{xx}^{\text{min}}| - \sigma_0$, and σ_0 is a φ -independent background. In both FM phases $\mathcal{R}$ is strongly field-dependent—it nearly ‘blows up’ at low fields in both. There are, however, significant differences. A closer look at the $\sigma_{yx}(\varphi)$ in FM 2 reveals deviations from a simple $\cos \varphi$ dependence (Fig. 3g) that are consistent with a new magnetic symmetry axis, see analysis in Supplementary Fig. 21.

Generally, since the Lorentz invariance in condensed matter and specifically in Weyl semimetals can be broken³⁷, Weyl nodes are permitted to tilt. In $\mathcal{T}$ -symmetry-broken (magnetic) Weyl semimetals the Weyl node partners with different chiralities tilt in the opposite directions, producing currents that are linear and antisymmetric functions of magnetic field^{14,33,38}. A Lorentz-violating quasiparticle is one whose momentum–energy relation depends on the direction it travels. To get a sense of the possible tilt directions, we take an in-depth look at the angular variations of the in-plane transport coefficients (Fig. 4). In FM 1, polar plots of $\sigma_{xx}^{\text{avg}}(\varphi)$ (Fig. 4b, c) and $\sigma_{yx}(\varphi)$ for

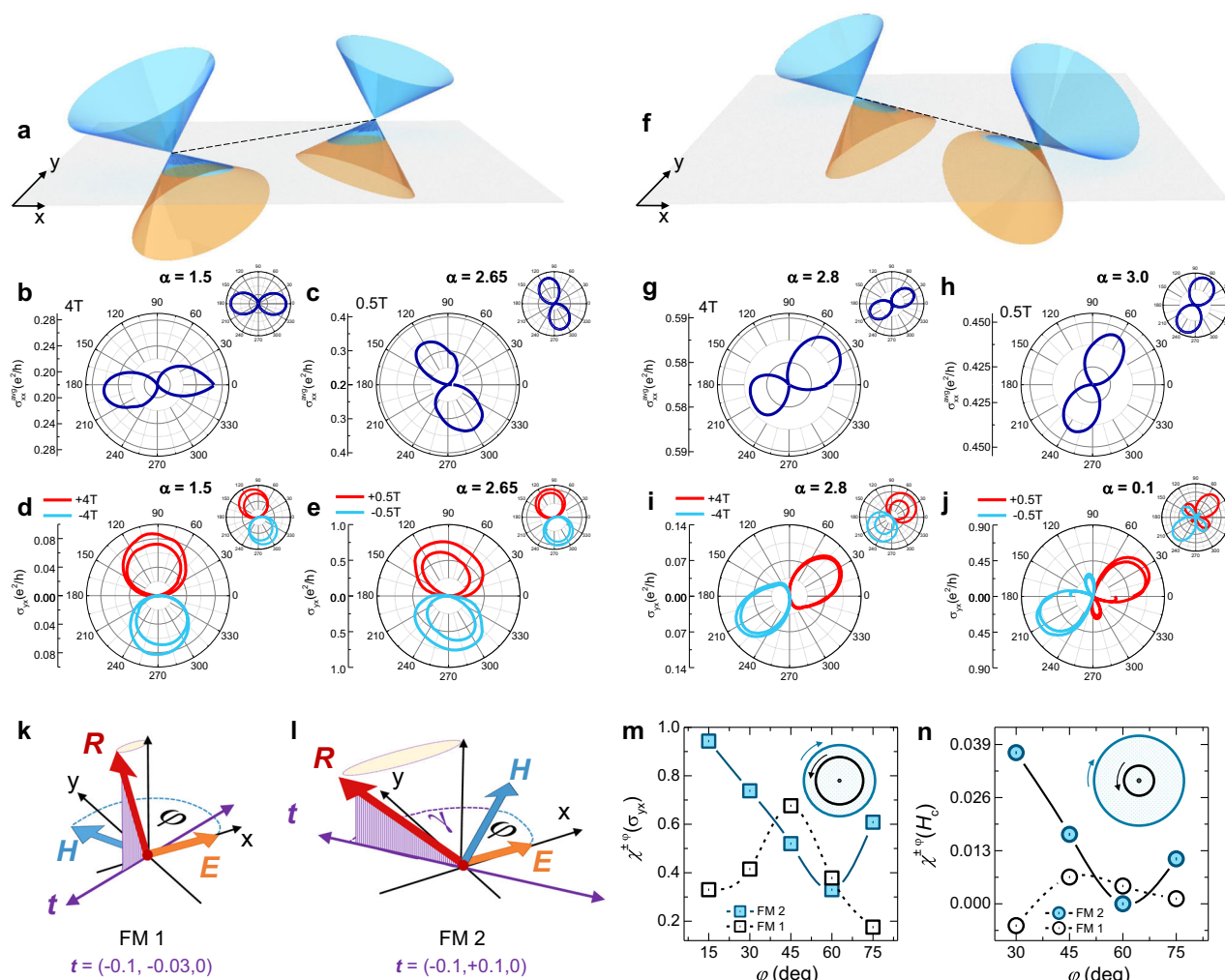


Fig. 4 | Symmetry change and Weyl cone tilt revealed under in-plane field rotation in hydrogen-modified MnSb_2Te_4 . **a** Illustration of energy-momentum dispersion for a pair of oppositely tilted Type I Weyl nodes representing FM 1. The position of the Fermi level in our p -type material is indicated as gray slab. Polar plots of the in-plane dependencies of $\sigma_{xx}^{avg}(\varphi)$ in FM 1: **b** at 4 T, and **c** at 0.5 T. The slanted ‘figure-eight’ at 0.5 T is accounted for by an increase in the scattering strength ratio α , labeled on the top (see Supplementary Note 20). Polar plots of $\sigma_{yx}(\varphi)$ in FM 1: **d**, at 4 T, and **e**, at 0.5 T, for positive (red) and negative (blue) fields. The values of α are in correspondence with $\sigma_{xx}(\varphi)$. **f** An energy-momentum dispersion for oppositely tilted and rotated Weyl nodes representing FM 2. In-plane polar plots of $\sigma_{xx}^{avg}(\varphi)$ in FM 2: **g**, at 4 T and **h**, at 0.5 T. In-plane polar plots of $\sigma_{yx}(\varphi)$ in FM 2 for positive and negative fields: **i** at 4 T and **j** at 0.5 T. **Top right insets:**

Theoretical polar plots (see Supplementary Figs. 24, 25) corresponding to data in (**b–e**) and (**g–j**). **k, l** Cartoons of three directed quantities: electric field $\vec{E}$, magnetic field $\vec{H}$, and the Weyl cone tilt direction $\vec{R}$ with its projection $\vec{t}$ in the xy -plane (see text). Tilt values obtained from theory are indicated. γ is the angle between tilt $\vec{t}$ and magnetic field. A much larger and rotated tilt $\vec{t}$ is represented in FM 2. **m, n** φ chirality $\chi^{\pm\varphi}$ of planar Hall signal $\sigma_{yx}(\varphi)$ (see text) and of the width H_c of the planar hysteresis loop $\chi^{\pm\varphi(H_c)} = H_c(\varphi) - H_c(-\varphi)$ in FM 1 and FM 2. Strong chirality amplification is seen in FM 2 outside of the characteristic angle (45–60°) range (see Supplementary Figs. 27, 28). For a subsequent hydrogenation cycle, see Supplementary Figs. 18, 19, and 30. The double circles in the insets illustrate the chiral circulation direction (arrows) and indicate circulation reversal in FM 2.

the positive and negative values of H (Fig. 4d, e) all have a ‘figure-eight’ shape with the elongated direction in σ_{xx}^{avg} rotated between high (4 T) and low (0.5 T) values of H from $\varphi = 0^\circ$ to 120° but remaining slightly off $\varphi = 90^\circ$ in σ_{yx} for all fields. These features are captured with good fidelity in a semiclassical theory of linear transport in Weyl semimetals when the ratio of internode/intranode scattering strength α is allowed to vary (Supplementary Note 20, Supplementary Figs. 24, 25).

Our starting point is a Type I Weyl³², consistent with the stoichiometry of our FM MnSb_2Te_4 (see the “Methods” section). By exploring the Weyl tilt and α parameter space of the model (see Supplementary Note 20) consistent with the experimental results, we deduce the Weyl tilt $\vec{t}$ in FM 1 to be of a moderately tilted Type I kind, nearly aligned with x -axis in the xy -plane (Fig. 4a). The parameter α here is nearly doubled from 1.5 at high H to 2.65 at a lower field, consistent with internode conductivity decreasing with increase in the magnetic field³⁶.

Discussion

Hydrogen-tuned transport chirality and reduced dissipation

The reorientation and modification of Weyl nodes in FM 2 seen in the polar plots (Fig. 4g–j) is striking. The figure-eights of σ_{xx} and σ_{yx} are now elongated along $\varphi = 30^\circ$ direction at high fields, corresponding to a larger tilt (Fig. 4f) and further rotation of $\vec{t}$ in the xy -plane. The φ -variations of σ_{xx} and σ_{yx} at low fields differ drastically—while the long axis of $\sigma_{xx}^{avg}(\varphi)$ points in the $\varphi = 60^\circ$ direction, corresponding (as in FM 1) to a higher α -value at low fields, $\sigma_{yx}(\varphi)$ has radically shape-changed to feature antisymmetry (Fig. 4j) with α reduced by an order-of-magnitude. Concurrently, there is a strong enhancement of net field-rotational chirality²⁰ of PHE $\chi^{\pm\varphi(H_c)} = \sigma_{yx}(\varphi) - \sigma_{yx}(-\varphi)$ (see Fig. 4m, n and Supplementary Figs. 27, 28), with φ -chiral PHE characterized by low internode dissipation prominent at low fields. Here net chirality $\chi^{\pm\varphi}$ is defined in the usual way, namely it is a lack of a coincident ‘mirror

image' in transport coefficients upon the in-plane angle φ switching between $\pm\varphi$.

Although theory fully describing the newly created Weyl phases is yet to be developed, the rational approach used above is a step forward to understanding the complex charge transport in a magnetic Weyl system. We infer that $\chi^{\pm\varphi}$ and low α attest to the chirality-guided PHE origins (see Supplementary Note 25)—associated with the consistently progressive tilt of the Weyl nodes (Supplementary Fig. 30)—of linear-in-current (Supplementary Fig. 31) out-of-plane-field antisymmetric magnetoresistance enforced by the hydrogen-altered tilt of the Weyl nodes of MST. And while our main experimental findings can be consistently understood in the framework of a Weyl model parameterized by the internode/intranode scattering ratio α , the nature of the sharp reduction of α in the planar Hall conductance and a possible constraint placed on this reduction by the in-plane $\sigma_{xx}(H)$ remains to be elucidated. For one, the role of Weyl orbits formed by the surface Fermi arcs and the chiral bulk channels⁴ and their contribution to σ_{xx} under tilted magnetic fields requires further theoretical and experimental investigation.

Lastly, we remark that our findings represent an advance in designer topological quantum materials. Tunable topological bandstructures facilitated by hydrogen or other light elements (e.g., lithium³⁹) through defect-related pathways expand the availability of robust and accessible platforms for exploring and harnessing topological phases with stunning macroscopic behaviors, opening a path to a potentially disruptive chirality-based implementations⁴⁰ in future quantum devices.

Methods

Crystal growth and structural characterization

Crystals of nominally MnSb_2Te_4 (MST) were grown out of a Sb–Te flux^{18,30}. Mixtures of Mn (Alfa Aesar, 99.99%), Bi and Sb pieces (Alfa Aesar, 99.999%), and Te shot (Alfa Aesar, 99.9999%) in the molar ratio of 1:10:16 ($\text{MnTe}:\text{Mn}_2\text{Te}_3 = 1:5$) were placed in a 2 ml alumina growth crucible and heated to 900 °C and held for 12 h. After slowly cooling across a ≈ 10 °C window below 600 °C for 2 weeks, the excess flux was removed by centrifugation above the melting temperature of Sb_2Te_3 (>620 °C). Crystals produced by this flux method were typically a few mm on a side and often grew in thick, block-like forms with thicknesses up to 2 mm but were easily delaminated.

- The samples for high-resolution transmission electron microscopy (TEM) investigations (Supplementary Fig. 1) were cut along the *c*-axis in the [1120] orientation, using a focused ion beam (FIB) method. TEM characterization was carried out in a FEI Talos F200X microscope operated at 200 kV. Structural observations were performed in scanning transmission electron microscope (STEM) mode using a high-angle annular dark field (HAADF) imaging. Energy dispersive X-ray spectroscopy (EDX) using a Super-X system with four silicon drift detectors (SDDs) was applied to detect differences in local chemical composition in the cross-sectional samples.
- Energy Dispersive X-ray (EDX) microanalysis was performed in the Zeiss Supra 55, a field emission SEM with a maximum resolution of 1 nm. The stoichiometry of the net ferromagnetic (possibly ferrimagnetic^{28,41}) bulk MST crystals used in this study was determined as corresponding to a somewhat Mn-deficient formula $\text{Mn}_{0.93}\text{Sb}_{2.07}\text{Te}_4$.
- X-ray diffraction of crystals was performed in a Panalytical diffractometer using Cu K α ($\lambda = 1.5405$ Å) line from Philips high intensity ceramic sealed tube (3 kW) X-ray source with a Soller slit (0.04 rad) incident and diffracted beam optics. The determined *c*-axis parameter = 40.898 Å was consistent with space group *R*-3m, as reported in the literature (see Supplementary Fig. 2).
- X-ray photoelectron spectroscopy (XPS) measurements (Supplementary Fig. 10) were performed in Physical Electronics

VersaProbeII XPS with a K α Al X-ray source (energy 1486.6 eV) and a hemispherical electron energy analyzer (pass energy 20 eV). The XPS energy-scale calibration was made with Bi 4f_{5/2} (162.5 eV), 4f_{7/2} (157.1 eV), Ag 3d_{5/2} (368.2 eV), and C 1s (284.8 eV) when available. PHI MultiPak software was used for XPS data reduction.

Hydrogenation protocol

Hydrogenation of exfoliated crystalline flakes and of already-fabricated devices was performed by submerging them in a dilute (0.5 M) HCl solution at room temperature for a predetermined period of time, then submerged and rinsed with deionized water and dried with N₂¹⁸. The diffusion of hydrogen H⁺ from the solution was controlled by the submerging time. Hydrogen was released from the samples by the vacuum anneals in the 60–200 °C range. The rate of this process is prescribed by the annealing temperature and the crystal thickness—at higher temperatures and/or for thinner crystals, the diffusion of hydrogen to the surfaces is faster.

Magnetic measurements

The d.c. magnetization measurements of the 0.01–0.02 mm-thick single crystal samples were performed in the superconducting quantum interference device (SQUID) magnetic property measurement system (MPMS). The system was calibrated using a Pd standard. Magnetization was measured up to 5 T and down to 2 K, with the sample accurately positioned in the pick-up coil by the centering scan dipole signal. From the fits to Curie–Weiss law, the magnetic moment in as-grown ferromagnetic MST crystals was determined as $\mu_{\text{eff}} = 5.36 \mu_{\text{B}}/\text{mole}$.

Transport measurements

Transport measurements were performed in a 14 T quantum design physical property measurement system (PPMS) in 1 Torr (at low temperature) of He gas on over ~ 30 samples, each subjected to the same in situ annealing protocol starting from 330 to 400 K during dehydrogenation. Crystals were mechanically exfoliated onto 300 nm SiO₂/Si⁺⁺⁺ wafers, typically resulting in 10–20 micron-size crystals with thicknesses less than 120 nm, as determined by the atomic force microscope (AFM). Electrical contacts in the van der Pauw (vdP) configuration⁴² were photo-lithographically patterned, and a sputtered Au metallurgy was used. Angle-resolved transport measurements with magnetic field rotated in- and out-of-plane were performed using vertical and horizontal rotator probes configured for 360° scans of both azimuthal φ and polar θ angles in the vdP geometry. To ensure consistent determination of net angular chirality $\chi^{\pm\varphi}$ across the entire field range while maintaining a constant M_z (the component of *M* along the *z*-axis), measurements were conducted at $\pm\varphi$, allowing for the polarity of the parallel field to be switched without altering M_z . To obtain the net chirality of PHE (distinct from the loop), the detected signal was symmetrized as described in the text. The vdP d.c. measurements were carried out using a custom-configured electronic system in which four measurement configurations are switched by a Keithley scanner, with the current direction reversal employed for each measurement to minimize thermal EMF. Sheet resistances were calculated using the vdP formulae, taking the vdP correction factor into account⁴³. Hydrogenation cycles were performed on the same pre-wired samples, thus eliminating any effects due to sample geometry and contacts. In all samples studied, the repeated hydrogenation cycles gave robust and reproducible results.

Data availability

The methods and data that support the findings in this work are available within the paper and in the Supplementary Information. Source data that support the plots within this paper are available at <https://doi.org/10.5281/zenodo.13748543>.

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Author contributions

Experiments were designed by L.K.-E. H.D. developed the hydrogenation protocol of as-grown materials. The materials modification and characterization, device fabrication and transport measurements were carried out by A.N.T., A.L., X.D., and E.B. HAADF-STEM and EDX elemental imaging was performed by K.S. DFT structures and electronic parameters of MnSb_2Te_4 under hydrogenation were calculated by K.P. Data analysis was done by A.N.T., E.B., X.D., A.L., and L.K.-E. G.S. and S.T. calculated planar transport coefficients of tilted Weyl cones from the Berry curvature. L.K.-E. wrote the manuscript with the input from all.

Competing interests

The authors declare no competing interests.

Additional information

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